

17. Option (d) is correct.

We know that if last two digit of a number divisible by 4 then number divisible by 4.

So, number is divisible by 4

$$= 1 \times 2 \times 3 \times 1 \\ = 1 \times 2 \times 3 \times 1 = 6$$

18. Option (a) is correct.

$$2^{120} = (2^3)^{40} = 8^{40} = (1+7)^{40} \\ = 1 + {}^{40}C_1 7 + {}^{40}C_2 7^2 + \dots + {}^{40}C_{40} 7^{40} \\ = 1 + 7[{}^{40}C_1 + {}^{40}C_2 7 + \dots + {}^{40}C_{40} 7^{39}]$$

$\therefore$ Remainder = 1

19. Option (c) is correct.

$$\begin{vmatrix} C(9,4) & C(9,3) & C(10,n-2) \\ C(11,6) & C(11,5) & C(12,n) \\ C(m,7) & C(m,6) & C(m+1,n+1) \end{vmatrix} = 0$$

Applying $C_3 \rightarrow C_1 + C_2 - C_3$

$$\begin{vmatrix} C(9,4) & C(9,3) & C(10,4) - C(10,n-2) \\ C(11,6) & C(11,5) & C(12,6) - C(12,n) \\ C(m,7) & C(m,6) & C(m+1,7) - C(m+1,n+1) \end{vmatrix} \\ = 0$$

Since determinant value is zero.

$$\therefore C_3 = 0$$

$$\text{So, } n-2 = 4$$

$$\Rightarrow n = 6$$

20. Option (b) is correct.

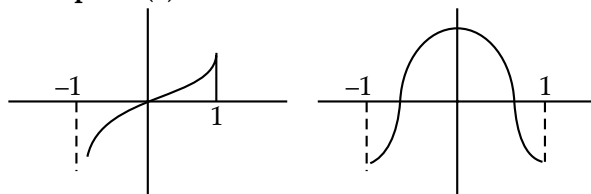
$$\begin{vmatrix} \cos C & \sin B & 0 \\ \tan A & 0 & \sin B \\ 0 & \tan(B+C) & \cos C \end{vmatrix} \\ = \cos C [-\sin B \tan(B+C)] - \tan A (\sin B \cos C) \\ = -\sin B \cos C \left[\frac{\sin(B+C)}{\cos(B+C)} + \frac{\sin A}{\cos A} \right] \\ = -\sin B \cos C \left[\frac{\sin(B+C) \cos A + \sin A \cos(B+C)}{\cos A \cos(B+C)} \right] \\ = -\sin B \cos C \frac{\sin(A+B+C)}{\cos A \cos(B+C)} \\ = \frac{-\sin B \cos C \sin(\pi)}{\cos A \cos(B+C)} \\ = 0$$

21. Option (d) is correct.

Possible order of 4 entries matrices

$$1 \times 4, 2 \times 2, 4 \times 1$$

$$\therefore \text{Total number of matrices} = 4 \times 4 \times 4 \times 4 \times 3 \\ = 768$$

22. Option (a) is correct.

$$f(x) = x|x|$$

is one-one and onto

$$g(x) = \cos(\pi x) \text{ is not}$$

one-one

23. Option (d) is correct.

$$\therefore xRy \Rightarrow |x+y| < 2 \text{ in } (-1, 1)$$

For reflexive

$$|x+x| < 2 \Rightarrow |x| < 1 \text{ true}$$

$\therefore$ R is reflexive

For symmetric

$$\text{Let } xRy \Rightarrow |x+y| < 2$$

$$\Rightarrow |y+x| < 2 \Rightarrow yRx$$

So, R is symmetric

For transitive

$$\therefore -1 < x < 1$$

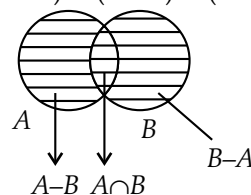
$$\text{Let } |x+y| < 2 \text{ and } |y+z| < 2$$

$$\text{then } |x+z| < 2$$

So R is transitive.

24. Option (a) is correct.

$$(A \cup B) - \{(A-B) \cup (B-A) \cup (A \cap B)\}$$



$$\therefore (A-B) \cup (B-A) \cup (A \cap B) = A \cup B$$

$$\therefore (A \cup B) - (A \cup B) = \phi \text{ (Null set)}$$

25. Option (a) is correct.

$$\begin{vmatrix} a^2 & b \sin A & c \sin A \\ b \sin A & 1 & \cos A \\ c \sin A & \cos A & 1 \end{vmatrix} \\ = a^2(1 - \cos^2 A) - b \sin A(b \sin A - c \cos A \sin A) \\ + c \sin A(b \sin A \cos A - c \sin A) \\ = a^2 \sin^2 A - b^2 \sin^2 A + bc \sin^2 A \cos A + bc \sin^2 A \cos A - c^2 \sin^2 A \\ = a^2 \sin^2 A - b^2 \sin^2 A - c^2 \sin^2 A + 2bc \sin^2 A \cos A \\ = \sin^2 A(a^2 - b^2 - c^2 + 2bc \cos A) \\ = \sin^2 A \times 0 = 0$$

26. Option (a) is correct.

$\therefore a, b, c$ are in AP

$$\Rightarrow 2b = a + c \quad \dots(i)$$

b, c, d are in GP

$$\Rightarrow c^2 = bd \quad \dots(ii)$$

c, d, e are in HP